

PLASMA SPRAY PROCESSORS Works Instruction	QF/QMS/02/020	
	Rev. 00	
	Date : 01/03/2019	

Operations/Work/Job Activity covered by this assessment: Contact Plate

Required Personal Protective Equipment (PPE) :



Safety glasses must be worn at times.



Long and loose hair must be contained.



Appropriate footwear with substantial uppers must be worn.



Close fitting/protective clothing to cover arms and legs must be worn.



Respiratory protection devices required.



Oil free leather gloves and spats must be worn.



Hearing protection must be used when using this machine.

Loading & Unloading Check:

1. Ensure no slip/trip is present in work pass and walk ways.
2. Ensure that work area is clean and clear of grease, oil and any flammable materials.
3. Refer Work Instructions PSP/WI/14 while handling of Contact Plate.

Operations Procedure:

- 1) Refer Pre-Inspection Work Instructions PSP/WI/03 to pre-inspect the Contact Plate.
- 2) If damaged is found on the coating surface, inform Gemsons Precision Engineering (P) Ltd. (GPEPL) by email with photograph and get their approval.
- 3) Repair to be carried w.r.t. (GPEPL) procedure.
- 4) Mark the area with the stencil w.r.t. drawing.
- 5) Mask with High Temperature Adhesive Tape/Metal Shim/Insulation tape to protect unwanted area from blasting and TWAC coating of Copper.
- 6) Ensure the Compressed Air is clean & free from water residues.
- 7) Refer Final Blasting Work Instructions PSP/WI/04 for preparing surface. The surface preparation will be as per Sa3 specs.
- 8) Use virgin 24mesh Al2O3 grit for blasting. Maintain Air pressure to blast at 3.5bar with 6mm bore nozzle and keeping 200 ±50mm standoff distances.
- 9) If any pinholes/micro pinholes are observed convey to supervisor for approval.
- 10) Maintain 5 to 8 Ra microns surface roughness during blasting and check visually.
- 11) Load the Contact Plate in the acoustic room and clamp it thoroughly on the fixture to prevent clamping marks on the Contact Plate.
- 12) Start Exhaust Fans before starting the TWAC Control Unit.
- 13) Open valve of Air and set the pressure according to the parameter mentioned in the Technical Bulletin.
- 14) Start the TWAC control unit and set Amps and Voltas desired in the parameter.
- 15) Closed the Acoustic Door and Wait for signal before igniting the TWAC Torch.

PLASMA SPRAY PROCESSORS Works Instruction	QF/QMS/02/020	
	Rev. 00	
	Date : 01/03/2019	

- 16) Press the start button to ignite the TWAC Torch and set the Amps by turning the Air Flow valve to get the desire Amps 0f 125 to 150 Amps.
- 17) Start the Robot for evenly spraying of Contact Plate.
- 18) Repeat the Cycle two times to achieve the thickness of 60 to 90microns.
- 19) Reverse the Contact Plate to spray the sides and repeat Step 16 to 18.

Shutdown :

1. Turn off the valve to Shut down the TWAC flame and bring the robot to its home position.

Inspection:

1. Check the dimension to get the 60+20 μ m thickness of the coating.
2. Brush the coated area with Stainless steel wire brush to remove loose particles from the coated surface.
3. Refer to Final Inspection Work Instruction PSP/WI/10 and get the approval of GPEPL Representative.

Assessor :	Prasanna Wadkar	Signature:		Date:	6/4/15
Approved:	Dhimant Doshi	Signature:		Date:	8/4/15